

IoT-Based Spectacle for Indoor Navigation and Safety of Visually Impaired

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Indoor navigation remains a major challenge for visually impaired individuals because GPS signals are unreliable in enclosed environments, and many traditional assistive methods do not provide accurate indoor position awareness. Although recent studies have introduced wearable and sensor-based navigation systems, many existing approaches still depend on fixed building infrastructure, continuous internet connectivity, or single sensor methods that limit positioning accuracy and practical usability. In addition, several assistive devices focus mainly on obstacle detection rather than continuous indoor self localization. This creates a research gap in developing a portable, reliable, and offline-capable indoor positioning solution that can support real-time indoor navigation for visually impaired users.

This research proposes a **Hybrid Indoor Positioning System with Offline Capability** designed specifically for indoor localization and navigation support of visually impaired users. The system integrates a camera module, IMU sensor, and offline map and landmark data within an embedded edge-computing architecture based on Raspberry Pi. The proposed approach combines vision-based landmark recognition, IMU-based motion tracking, and sensor fusion to estimate the user's indoor position and heading in real time. By performing localization and processing locally on the embedded device, the system is intended to operate without GPS and without continuous internet connectivity, while still providing reliable position output for guidance support.

The expected outcome is a portable, cost-effective, and scalable assistive indoor positioning component that improves localization reliability compared to single-sensor or infrastructure-dependent systems. Socially, the proposed solution aims to enhance independence, mobility confidence, and safety for visually impaired individuals in indoor environments such as hospitals, universities, and public buildings. Technically, it contributes to assistive technology research by introducing an offline, hybrid, and edge-based indoor positioning architecture that can be extended for broader smart navigation applications.

Keywords: Hybrid Indoor Positioning System; Offline Capability; Indoor Navigation; Sensor Fusion; Assistive Technology

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
BLE	Bluetooth Low Energy
EKF	Extended Kalman Filter
ERC	Ethics Review Committee
GPS	Global Positioning System
HCI	Human-Computer Interaction
IMU	Inertial Measurement Unit
IoT	Internet of Things
LiDAR	Light Detection and Ranging
LKR	Sri Lankan Rupee
NFC	Near Field Communication
NGO	Non-Governmental Organization
RFID	Radio Frequency Identification
RGB-D	Red, Green, Blue – Depth
SDG	Sustainable Development Goal
SLAM	Simultaneous Localization and Mapping
SLIIT	Sri Lanka Institute of Information Technology
UWB	Ultra-Wideband
WBS	Work Breakdown Structure
WHO	World Health Organization
YOLO	You Only Look Once (Object Detection Algorithm)

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1. INTRODUCTION

Indoor navigation is one of the most significant challenges in assistive technology for visually impaired users because conventional GPS-based navigation systems do not function reliably inside buildings. Unlike outdoor environments, indoor spaces require highly accurate, context-aware, and continuous localization to help users safely identify corridors, rooms, stairs, elevators, and nearby obstacles. Existing research shows that visually impaired users need indoor navigation systems that are more precise, responsive, and reliable than general-purpose navigation tools, which has led researchers to explore alternatives such as inertial sensing, computer vision, Wi-Fi, RFID, BLE, and hybrid localization approaches [1], [3], [7].

Several studies demonstrate that combining multiple sensing sources provides better indoor positioning performance than relying on a single technology. Gomes et al. proposed an indoor navigation architecture that adapts to different environmental data sources, while Hesch and Roumeliotis showed that portable indoor localization can be achieved through multi-sensor fusion without depending on GPS [1], [3]. In addition, studies such as Blind Shopping illustrate that smartphone-based indoor guidance can support visually impaired users effectively, although such systems often rely on fixed infrastructure such as RFID tags and store-specific setups [2], [4]. A broader review by Plikynas et al. further emphasizes that hybrid and camera-based indoor navigation systems are more flexible and better aligned with user needs because they can reduce dependency on installed infrastructure while improving localization and orientation accuracy [7].

More recent research has strengthened the case for wearable and locally processed navigation systems. Chen et al. introduced a wearable navigation device based on real-time semantic visual SLAM, showing that indoor localization, mapping, and environmental understanding can be achieved on an embedded platform for visually impaired users [8]. Similarly, other wearable assistive systems using embedded devices such as Raspberry Pi have shown that real-time sensing, object detection, and local computation are practical for assistive mobility applications [5], [9]. Research on low-cost edge devices also supports the feasibility of running real-time sensor-fusion and estimation tasks offline, which is important for portable assistive systems that must operate without constant internet connectivity [6].

Based on these findings, the Hybrid Indoor Positioning System with Offline Capability is proposed as a core component of the smart assistive platform. This component is designed to estimate the user's position inside buildings by combining complementary sensing techniques, such as camera-based landmark recognition and IMU-based movement tracking, in order to produce more stable and accurate indoor localization. The offline capability of the component is especially important because many existing assistive solutions depend on cloud services, internet access, or building-specific infrastructure, which can reduce portability, privacy, and reliability in real-world usage [1], [7], [8].

Ultimately, this component acts as the indoor navigation foundation of the overall system. By continuously identifying the user's indoor position and movement direction without relying on GPS or constant network connectivity, it enables higher-level modules such as hazard detection, route assistance, and audio feedback to function effectively. Therefore, the Hybrid Indoor Positioning System with Offline Capability is not only a positioning module, but also a critical enabler of safe, intelligent, and independent indoor mobility for visually impaired individuals [5], [7], [8], [9].

1.1 Background and Context

Indoor navigation is still a serious challenge for visually impaired people because GPS does not work well inside buildings. In places such as hospitals, universities, malls, and offices, users often struggle to identify rooms, corridors, stairs, and exits safely. This lack of reliable indoor positioning affects confidence, safety, and independent movement, making it an important accessibility problem rather than only a technical issue [1], [3], [7], [8].

This problem mainly affects blind and low-vision individuals, but it also impacts caregivers, families, educators, employers, and service providers who often need to assist them. Indoor mobility is important because it directly affects independence, access to education, healthcare, work, and daily activities. Without proper navigation support, visually impaired users may depend on others even for simple tasks such as finding a classroom or hospital unit [7], [8], [13], [16].

The issue is significant worldwide and in Sri Lanka. According to WHO, at least 2.2 billion people live with vision impairment globally, while around 1.3 billion people experience significant disability [12], [13]. In Sri Lanka, census data show that 996,939 people had difficulties related to seeing, and national studies reported 1.7% blindness and 17% low vision among adults over 40 [11], [14]. These numbers show that accessible indoor navigation is a highly relevant need both globally and locally [11], [12], [13], [14].

Many existing indoor navigation systems rely on RFID, BLE, Wi-Fi, QR codes, or other building infrastructure. While these can work in controlled environments, they are often costly, location-specific, and difficult to scale [2], [4]. Other systems use sensors such as gyroscopes, laser scanners, or cameras, but they may depend on maps, special equipment, or high processing power [3], [7], [8]. Some smart wearable solutions focus only on obstacle detection and do not provide full indoor position tracking, which leaves an important gap [5], [7], [8], [9].

To overcome these issues, this research proposes a Hybrid Indoor Positioning System with Offline Capability. The idea is to combine several methods to improve indoor localization accuracy and reduce dependence on external infrastructure. The system includes a camera-based landmark recognition module, an IMU-based motion tracking module, a sensor fusion engine, an offline map and routing layer, and an edge-processing unit running on a device such as Raspberry Pi. Together, these parts aim to provide accurate, portable, and real-time indoor navigation without requiring internet access [1], [3], [5], [7], [8].

This component supports several Sustainable Development Goals. It aligns with SDG 3 by improving safe mobility and well-being, SDG 4 by supporting access to education, and SDG 8 by helping visually impaired individuals access workplaces more independently [13], [16], [17]. It also connects with SDG 10 by reducing inequality and SDG 11 by contributing to more inclusive and accessible built environments [16]. Therefore, the proposed solution has both technical and social value.

This research fits into multiple research areas. It belongs to Artificial Intelligence and Computer Vision because it uses visual recognition for localization [7], [8]. It also relates to IoT, Embedded Systems, and Wearable Computing since it is designed for portable hardware with local processing [5], [6], [9]. In addition, it connects with Assistive Technology and Human-Computer Interaction because it focuses on user centered navigation support, and with Robotics and Sensor Fusion because it combines motion sensing, mapping, and localization techniques [3], [8], [10].

1.2 Literature Review

Recent research on indoor navigation for blind and visually impaired users has moved from simple infrastructure-based methods toward more intelligent, wearable, and sensor-fusion-based systems. A major trend is the shift from single-method approaches, such as RFID-only or Wi-Fi-only positioning, to **hybrid indoor localization systems** that combine cameras, inertial sensors, depth sensing, and embedded processing for better accuracy and usability [1], [3], [7], [8]. Studies also show increasing interest in systems that can work with less building infrastructure and more local processing, which is especially important for practical assistive navigation in indoor environments [5], [7], [8].

• Hybrid indoor positioning and localization approaches

Hybrid indoor positioning is one of the most important research directions in this area. Gomes et al. proposed an indoor navigation architecture that combines multiple data sources such as Wi-Fi, BLE, visual tags, NFC, and inertial sensing to improve indoor positioning flexibility [1]. Hesch and Roumeliotis presented a portable indoor localization aid that uses a gyroscope, laser scanner, pedometer, and building map to estimate user position and heading indoors [3]. More recently, Chen et al. introduced a wearable navigation device based on semantic visual SLAM and an RGB-D camera, showing that real-time localization and mapping can be achieved on a wearable platform [8]. In addition, Plikynas et al. reviewed indoor navigation systems for visually impaired users and highlighted that hybrid and camera-based systems are increasingly preferred because they are more adaptable and better aligned with user needs [7].

- **Embedded/edge processing and offline assistive navigation**

Another important area is the use of embedded and edge devices for real-time processing. Chen et al. developed a wearable assistive system using object detection, stereo distance measurement, and tactile presentation on Raspberry Pi-based hardware, demonstrating that assistive perception tasks can be handled locally on portable devices [5]. García Guzmán et al. also showed that real-time sensor fusion and neural-network-based estimation can be performed on low-cost devices such as Raspberry Pi, supporting the feasibility of offline edge computation [6]. These studies are important because they support the idea that indoor navigation systems should reduce dependence on cloud services and internet connectivity while maintaining realtime responsiveness [5], [6].

- **Wearable smart-assistive systems**

Research has also expanded toward wearable assistive systems that combine sensing and user feedback. Chen et al. proposed a wearable navigation device that provides voice guidance using semantic scene understanding and real-time visual localization [8]. Swathi et al. presented IoT-based smart glasses that integrate a Raspberry Pi, camera, ultrasonic sensing, and audio feedback for obstacle awareness and basic navigation support [9]. In addition, Chen et al. developed a wearable tactile assistive system that combines object detection and distance measurement to improve situational awareness for visually impaired users [5]. These studies show that wearable form factors are becoming more practical, but they also reveal that many systems focus more on obstacle detection and scene awareness than on full indoor self-positioning.

- **Infrastructure-based navigation systems**

Earlier and still widely studied approaches rely on building-installed infrastructure for navigation. Blind Shopping, presented by López-de-Ipiña et al., uses RFID tags on the floor, a cane-mounted reader, and a smartphone to guide visually impaired users through supermarkets [2], [4]. These systems demonstrate that indoor navigation assistance can be effective in structured environments, especially when combined with audio instructions and mobile interfaces [2], [4]. However, they depend heavily on preinstalled RFID tags, QR codes, and environment-specific setup, making them less flexible for general-purpose indoor navigation across different buildings [2], [4], [7].

- **Existing products and services**

Existing systems and services show that assistive indoor navigation is becoming more wearable and user-centered, but most solutions are still limited in scope. Blind Shopping can be viewed as a service-oriented navigation system for supermarket environments using smartphone-based verbal guidance [2], [4]. The portable localization aid by Hesch and

Roumeliotis functions as a wearable navigation support system that estimates indoor position using a cane-mounted sensor setup [3]. Chen et al.'s semantic visual SLAM device and tactile assistive system also reflect the growing move toward smart wearable solutions with real-time feedback [5], [8]. Swathi et al.'s smart glasses further show the interest in integrating multiple functions such as object detection, distance sensing, and audio guidance into one wearable platform [9]. These systems indicate strong progress, but most are still research-driven prototypes or environment-specific solutions rather than complete, general-purpose indoor positioning products.

- **Limitations of existing systems**

Although the current studies make valuable contributions, several limitations remain. Infrastructure-based systems such as RFID or BLE navigation depend on buildingspecific installation and maintenance, which reduces portability and scalability [2], [4], [7]. Portable localization systems can improve accuracy, but many rely on special hardware, pre-known maps, or cane-mounted equipment that may not be convenient for everyday wearable use [3]. Vision-based and SLAM-based systems provide richer localization and scene understanding, but they can be affected by lighting, occlusion, computation cost, and battery limitations [7], [8]. In addition, many wearable assistive devices focus mainly on obstacle detection or object recognition rather than continuous indoor position estimation [5], [9]. These limitations show the need for a **Hybrid Indoor Positioning System with Offline Capability** that can combine multiple sensing methods, reduce reliance on infrastructure, and support real-time indoor localization through local embedded processing [1], [7], [8].

1.3 Research Gap

Although there has been significant progress in assistive indoor navigation for visually impaired users, existing systems still have several important limitations. Most current studies focus only on specific parts of the problem, such as obstacle detection, infrastructure-based localization, or scene recognition, rather than providing a **complete, portable, and reliable indoor positioning solution** for real-world use. Based on the literature review, it is clear that there is still a lack of a **hybrid indoor positioning system with offline capability** that can accurately localize a user indoors without depending heavily on internet access or building-installed infrastructure [1], [3], [7], [8].

Based on the literature review, the following research gaps have been identified:

- **Dependence on building infrastructure**

Many existing indoor navigation systems rely on RFID tags, BLE beacons, Wi-Fi fingerprints, QR markers, or other pre-installed building infrastructure. While these systems can work in controlled environments, they are costly to deploy, difficult to maintain, and not suitable for general use across different indoor spaces [1], [2], [4], [7].

- **Limited hybrid sensor integration**

Several existing systems use only one main sensing method, such as RFID, camera vision, or motion tracking, which reduces reliability in complex indoor conditions. Effective indoor localization requires better integration of complementary sensing methods such as vision-based landmark recognition and IMU-based movement tracking, but this is still not fully achieved in many current solutions [3], [7], [8].

- **Weak offline capability**

Some assistive systems depend on cloud services, servers, or internet connectivity for processing and guidance. This can introduce latency and reduce system reliability in places where network access is poor or unavailable. For visually impaired users, indoor navigation must remain available in real time even without constant connectivity [5], [6], [8], [9].

- **Incomplete indoor self-localization**

A number of wearable assistive devices focus mainly on obstacle detection, object recognition, or audio feedback rather than continuously estimating the user's indoor position. As a result, many systems can warn users about nearby hazards but cannot reliably determine where the user is inside the building or support route-level navigation [5], [8], [9].

- **Limited practicality for everyday wearable use**

Some proposed systems use cane-mounted devices, complex sensor setups, or bulky hardware arrangements that may reduce comfort and everyday usability. For real-world adoption, indoor positioning support should be lightweight, portable, and suitable for continuous use in common indoor environments [3], [5], [8].

Due to these limitations, there is a clear need for an indoor navigation component that is **portable, infrastructure-light, sensor-fused, and capable of operating offline**. Existing approaches do not fully combine these qualities in a single practical solution for visually impaired users.

Therefore, this research proposes a **Hybrid Indoor Positioning System with Offline Capability**, which combines **vision-based landmark recognition, IMU-based motion tracking, sensor fusion, and edge processing** within a single assistive module. By performing localization and navigation support on an embedded device such as Raspberry Pi, the proposed system aims to improve indoor position estimation, reduce dependence on building infrastructure, and provide reliable real-time operation without requiring continuous internet access.

The proposed component contributes to the field of assistive technology by introducing a more **practical, privacy-preserving, and wearable indoor positioning solution** designed to improve independent mobility for visually impaired individuals in real-world indoor environments.

2. OBJECTIVE

2.1 Main Objective

To **design, develop, and evaluate a Hybrid Indoor Positioning System with Offline Capability** that can estimate the indoor location of visually impaired users in real time with an accuracy of **within 1–2 meters**, while operating on a local embedded device **without continuous internet connectivity**.

1. To develop a vision-based landmark recognition module

To design and implement a camera-based module that can detect important indoor landmarks such as doors, corridors, staircases, and room signs with at least **85% recognition accuracy** in a controlled indoor environment.

2. To implement an IMU-based motion tracking module

To develop a motion-tracking component using IMU sensors to identify user steps, turns, and movement direction with at least **80% tracking consistency** during indoor navigation tests.

3. To build a hybrid sensor-fusion localization engine

To integrate visual landmark data and IMU motion data into a single localization model that improves indoor positioning accuracy compared with using only one sensing method.

4. To enable offline indoor positioning and route support

To develop a local processing and offline map layer that allows the system to continue positioning and navigation assistance **without active internet access** during testing.

5. To deploy the system on an embedded edge platform

To implement the proposed indoor positioning component on a portable device such as **Raspberry Pi** and achieve localization updates with a response time of **less than 2 seconds**.

6. To evaluate the effectiveness of the proposed system

To test the prototype in selected indoor environments and measure its performance based on **positioning accuracy, response time, offline usability, and reliability** for visually impaired user navigation.

2.2 Specific Objectives

- To design and implement a camera-based module that can identify key indoor landmarks such as doors, corridors, staircases, and room signs with at least **85% detection accuracy** in controlled indoor test conditions.
- To develop an IMU-based movement tracking module that can detect user steps, turns, and walking direction with at least **80% tracking consistency** during indoor navigation trials.
- To integrate visual landmark data and IMU motion data into a hybrid localization model that can estimate the user's indoor position with an accuracy of **within 1– 2 meters** in selected indoor environments.
- To develop a local map and processing layer that allows the system to continue indoor positioning and basic route support **without internet connectivity** during experimental testing.
- To implement the proposed component on a portable platform such as **Raspberry Pi** and achieve localization updates with a processing response time of **less than 2 seconds**.
- To test the prototype in selected indoor locations and assess its performance using measurable criteria such as **position accuracy, response time, offline usability, and reliability** for visually impaired user navigation.

2. METHODOLOGY

3.1 Research Approach

This research is conducted under the **Design Science Research** approach, where the main aim is to design and develop a technological artifact that solves a real-world problem. The technological artifact proposed in this research is a **Hybrid Indoor Positioning System with Offline Capability**, which is intended to help visually impaired users determine their location and movement inside indoor environments without depending on GPS or continuous internet connectivity.

In addition, this research adopts an **experimental evaluation approach**, where the effectiveness of the proposed system will be tested in controlled indoor environments. The prototype developed in this research will be evaluated by measuring how accurately it can recognize indoor landmarks, track user movement, and estimate position in real time. The experimental setup will help determine whether the proposed hybrid and offline-capable approach is suitable for assistive indoor navigation.

3.2 Technical Methods and Algorithms

The proposed system uses several technical methods for indoor positioning and movement tracking.

The **camera module** captures real-time visual data from the surrounding indoor environment. Computer vision techniques will be used to identify indoor landmarks such as doors, corridors, staircases, and room signs. These landmarks will help the system estimate the user's location inside the building.

The **IMU sensor** is used to capture motion data such as acceleration, turning direction, and user movement. This supports **dead reckoning**, where the system estimates the user's position based on motion history such as steps and changes in direction.

The outputs from the camera and IMU will be combined using a **sensor fusion method**, such as an **Extended Kalman Filter (EKF)** or a weighted fusion approach. This helps improve localization accuracy by reducing drift from the IMU and correcting position estimates using visual landmarks.

An **offline map and route layer** will also be included so that basic building information and route references can be stored locally on the device. This allows the system to continue functioning without active internet connectivity.

All sensor data and localization logic will be processed on a **Raspberry Pi edge computing platform**, enabling real-time indoor positioning on a wearable or portable embedded system.

3.3 Tools and Platforms with Justification

The development of the proposed system requires several hardware and software tools.

- **Raspberry Pi 4 Model B**

The **Raspberry Pi 4 Model B** will be used as the main processing unit of the proposed system. This is because it provides better computational capability for computer vision, sensor fusion, and offline local processing while still being suitable for portable embedded deployment.

- **Raspberry Pi Camera Module**

The **Raspberry Pi Camera Module** will be used to capture real-time visual data from the indoor environment. The camera input will be used for recognizing landmarks such as doors, corridors, staircases, and other navigation-related features.

- **MPU6050 IMU Sensor**

The **MPU6050 IMU sensor** will be used to detect user movement, turning direction, and motion changes. This sensor supports dead reckoning and improves indoor localization when visual landmark data is temporarily unavailable.

- **Local Storage / Offline Map Data**

A lightweight local storage method such as **JSON files or SQLite** will be used to keep map-related data, landmark references, and route information. This supports the offline functionality of the system.

- **Rechargeable Battery Power Source**

A rechargeable battery pack will be used to make the system portable and wearable. This allows the prototype to function as an independent assistive module without relying on a fixed power source.

- **Software Tools (Python, OpenCV, NumPy)**

For software development, **Python** will be used as the main programming language. **OpenCV** will support computer vision and landmark recognition, while **NumPy** and related libraries will be used for sensor data handling and mathematical processing. These tools are selected because they are suitable for rapid development, hardware integration, and real-time embedded applications.

3.4 Data Collection and Ethical Considerations

The data collection for this research will focus on indoor landmarks, motion patterns, and localization data found in typical indoor environments. The data will be collected from locations such as hallways, rooms, staircases, junctions, and corridors.

The data used in the research will include images of indoor landmarks, IMU motion readings, walking paths, turning points, and reference locations inside the experimental environment. These data will be used to test the landmark recognition module, movement tracking module, and the combined localization performance of the proposed system.

The data used in this research may include images captured during experimentation and sensor readings collected directly from the prototype device. Ground reference points in the indoor environment will be marked to compare the estimated position with the actual location during testing.

Ethical Considerations

This research does not aim to collect personal or sensitive data from individuals. When image data is captured, the focus will be on environmental features such as doors, walls, corridors, and staircases rather than personal identity. If any person appears in the captured images during experiments, their identity will not be stored or used.

If human participants are involved in testing, informed consent will be obtained before experimentation. All collected data will be used only for research purposes, and the testing process will be conducted in a way that protects privacy, safety, and dignity. Special attention will be given to safe indoor testing conditions if visually impaired users participate in the evaluation.

3.5 Validation Strategy and Performance Metrics

The performance of the proposed system will be evaluated through experimental testing in indoor environments using predefined routes and known reference points. The estimated positions generated by the system will be compared with actual reference locations to determine localization accuracy.

The following performance metrics will be used to evaluate the system:

- **Landmark recognition accuracy** – the ability of the system to correctly identify indoor landmarks such as doors, corridors, staircases, and room signs.
- **Positioning accuracy** – the ability of the system to estimate the user's location correctly, measured in meters from the actual reference position.
- **Motion tracking consistency** – the reliability of the IMU-based module in detecting steps, turns, and direction changes during movement.
- **Response time** – the time taken for the system to process sensor input and update the user's estimated indoor position.
- **Offline operation success rate** – the ability of the system to continue functioning correctly without internet connectivity.
- **Localization reliability** – the stability and continuity of position estimation while the user moves through indoor routes.

The experimental results will be analyzed to determine whether the proposed hybrid approach improves indoor positioning performance compared to relying on a single sensing method. The results will also be used to evaluate whether the system is suitable as a practical assistive indoor positioning component for visually impaired users.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution

The proposed solution is a **Hybrid Indoor Positioning System with Offline Capability** that helps visually impaired users identify their location and movement inside buildings. The system collects **visual data** and **motion data** using a camera and IMU sensor, processes the data on an edge computing unit (Raspberry Pi), and estimates the user's indoor position in real time. The system is designed to work **without GPS** and can continue functioning **without continuous internet connectivity** by using locally stored maps and landmark data.

4.2 System Components

A) Hybrid Indoor Positioning Hardware Module

- **Processing Unit:** Raspberry Pi 4 Model B
Used as the main edge computing platform for local processing and sensor fusion.
- **Vision Sensor:** Raspberry Pi Camera Module
Captures real-time indoor visual data for landmark recognition.
- **Motion Sensor:** MPU6050 IMU Sensor
Detects movement, turns, and direction changes for motion tracking.
- **Storage:** MicroSD card
Stores the operating system, program files, local map data, and logs.
- **Power Unit:** Rechargeable battery / power bank
Provides portable power for the embedded system.

B) Software / Processing Layer

- **Image capture and preprocessing**
Captures and prepare visual data from the camera.
- **Landmark recognition module**
Detects indoor features such as doors, corridors, staircases, and room signs.
- **IMU-based motion tracking**
Tracks movement, turns, and walking direction.
- **Sensor fusion and localization engine**
Combines camera and IMU data to estimate indoor position.
- **Offline map and route layer**
Stores local landmark and building data for offline use.
- **Position output interface**
Sends location and heading information to the next assistive module.

4.3 High Level System Diagram (Components)

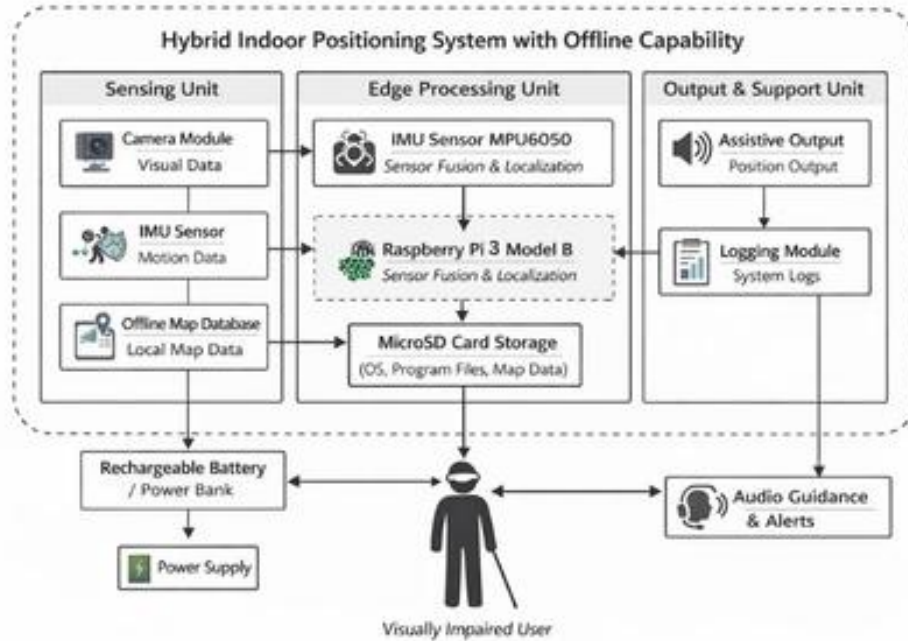


Figure 1 : High-Level System Architecture

4.4 Component Interactions and Data Flow (Hybrid Indoor Positioning System with Offline Capability)

The **Hybrid Indoor Positioning System** can be divided into three main functional sub-units:

1. Sensing Unit

This unit is responsible for collecting visual and motion data from the indoor environment.

- **Camera Module** – captures indoor visual scenes for landmark recognition
- **MPU6050 IMU Sensor** – detects user movement, turns, and orientation changes
- **Offline Map / Landmark Data** – provides locally stored environment references

2. Edge Processing Unit

This unit performs real-time localization and decision-making.

- **Raspberry Pi 4 Model B** – acts as the main edge computing device
- **Landmark Recognition Module** – identifies doors, corridors, stairs, and room signs
- **Motion Tracking Module** – estimates movement using IMU data
- **Sensor Fusion and Localization Engine** – combines visual and motion data to estimate indoor position and heading

3. Output and Support Unit

This unit provides the processed indoor position for navigation support and system evaluation.

- **Position Output Interface** – sends position and heading data to the next assistive module
- **Offline Map Layer** – supports route and landmark referencing without internet
- **Logging Module** – stores localization results and test logs

4.5 Scalability and Security Considerations

- **Camera → Raspberry Pi:** Sends image frames for landmark recognition and visual analysis.
- **IMU → Raspberry Pi:** Sends motion and orientation data for movement tracking.
- **Offline Map Database → Raspberry Pi:** Provides local landmark and building references for localization correction.
- **Raspberry Pi → Localization Engine:** Processes combine visual and IMU data to estimate indoor position.
- **Localization Engine → Output Interface:** Sends current position, heading, and confidence information to the next assistive module.
- **Battery → All Components:** Powers the system and supports portable operation.

4.6 Data Flow Explanation

1. **Capture:** The camera continuously captures indoor images.
2. **Sense motion:** The IMU records movement, direction changes, and orientation.
3. **Reference locally:** The offline map database provides stored landmark and route information.
4. **Process on edge:** The Raspberry Pi performs:
 - image preprocessing
 - landmark recognition
 - IMU-based motion tracking
 - sensor fusion
 - indoor position estimation
5. **Decision:** The system determines the user's current indoor position and heading.
6. **Output:** The estimated position is sent to the guidance or assistive output module for navigation support.
7. **Log:** Position results and system performance data are stored for testing and evaluation.

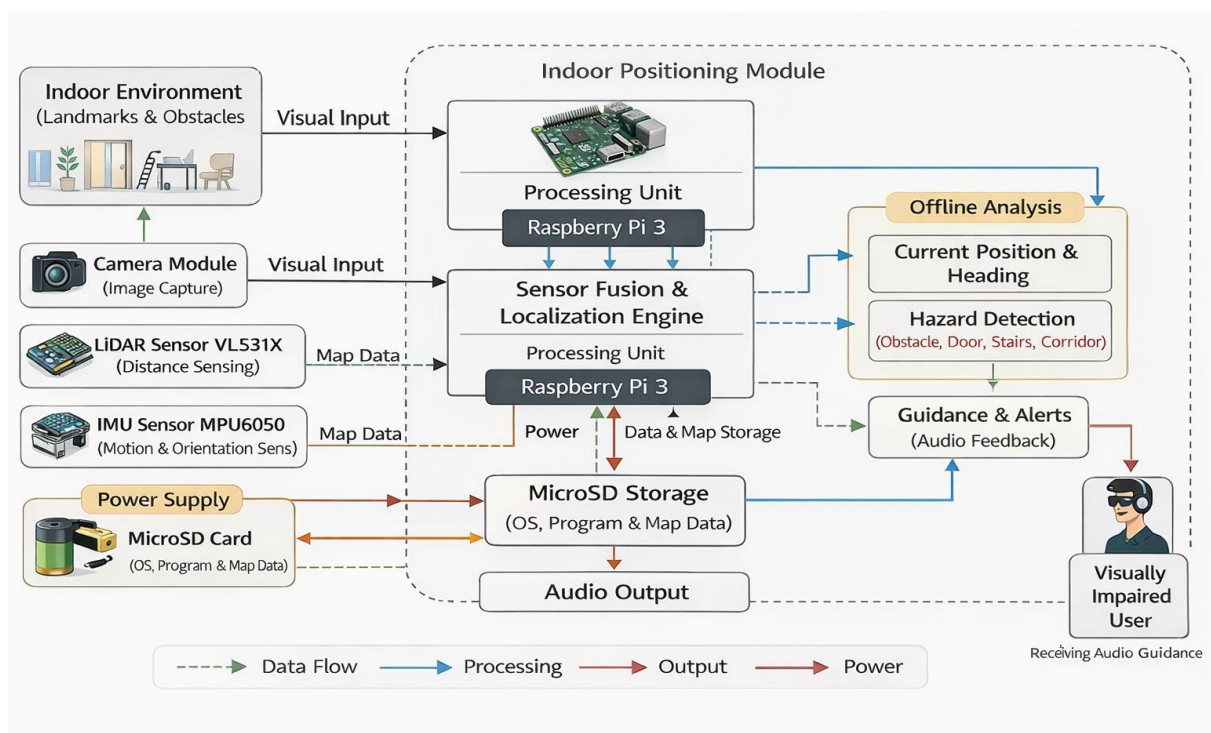


Figure 2 : Data Flow of the Proposed Smart Spectacle System

4.7 Scalability and Security Considerations

The proposed indoor positioning system is designed with scalability in mind so that additional assistive features can be integrated in the future without major changes to the architecture. Since the system uses a modular hardware and software design with edge computing, new functions can be added to improve indoor navigation support for visually impaired users.

Scalability

Possible future scalability improvements include:

- **Additional sensor integration:** The system can be extended with sensors such as BLE, UWB, or depth sensors to improve localization accuracy.
- **Multi-building support:** The system can be improved to store and use multiple indoor maps and landmark datasets for different buildings.
- **Connected assistive modules:** The system can later be integrated with audio guidance, mobile applications, or guardian support systems.

Security Considerations

Security and privacy are important in the development of assistive indoor positioning systems.

- **Privacy protection:** The system focuses on environmental landmark recognition and position estimation. Captured data should not store unnecessary personal identity information.
- **Offline processing:** The proposed system performs its main processing locally on the Raspberry Pi edge computing platform. This reduces the need to transmit sensitive data over the internet.
- **Secure data handling:** Any stored logs, map data, or system outputs should be used only for research and system operation, with restricted access to prevent misuse.
- **User safety:** The system should provide stable and reliable indoor position estimation so that visually impaired users can receive safer navigation support in indoor environments.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

- The **Hybrid Indoor Positioning System with Offline Capability** has several stakeholders who will be affected by the system.
- **Visually Impaired Users:** These are the primary users of the system. They will use the proposed component to identify their position, direction, and movement inside buildings without depending on GPS or continuous internet connectivity.
- **Researchers and Developers:** The researchers and developers of the proposed system are key stakeholders because they will design, implement, test, and improve the indoor positioning component.
- **Caregivers and Family Members:** Caregivers and family members of visually impaired individuals are also stakeholders, since safer and more reliable indoor positioning can improve user independence and reduce the need for constant assistance.
- **Assistive Technology Organizations:** Organizations that support accessibility and assistive technology may become future adopters or promoters of the proposed system in educational, healthcare, or public environments.
- **Institutions and Indoor Facility Providers:** Places such as universities, hospitals, offices, and public buildings are also stakeholders because the proposed system can be used in their indoor spaces to improve accessibility for visually impaired users.

5.2 Functional Requirements

The functional requirements define the main functions that must be performed by the system.

FR1: The system should be able to capture real-time visual data through the camera module.

FR2: The system should be able to collect motion and orientation data through the IMU sensor.

FR3: The system shall recognize indoor landmarks such as doors, corridors, staircases, junctions, and room signs from captured visual data.

FR4: The system shall track user movement, direction changes, and turning patterns using IMU data.

FR5: The system shall combine camera-based landmark recognition and IMU-based motion tracking to estimate the user's indoor position and heading.

FR6: The system shall store and access local map or landmark reference data for offline use.

FR7: The system shall process sensor data in real time using the Raspberry Pi edge computing platform.

FR8: The system shall continue indoor positioning functions without requiring continuous internet connectivity.

FR9: The system shall generate position-related output such as current location, direction, and confidence level for use by other assistive modules.

FR10: The system shall record localization results and performance logs for testing and evaluation purposes.

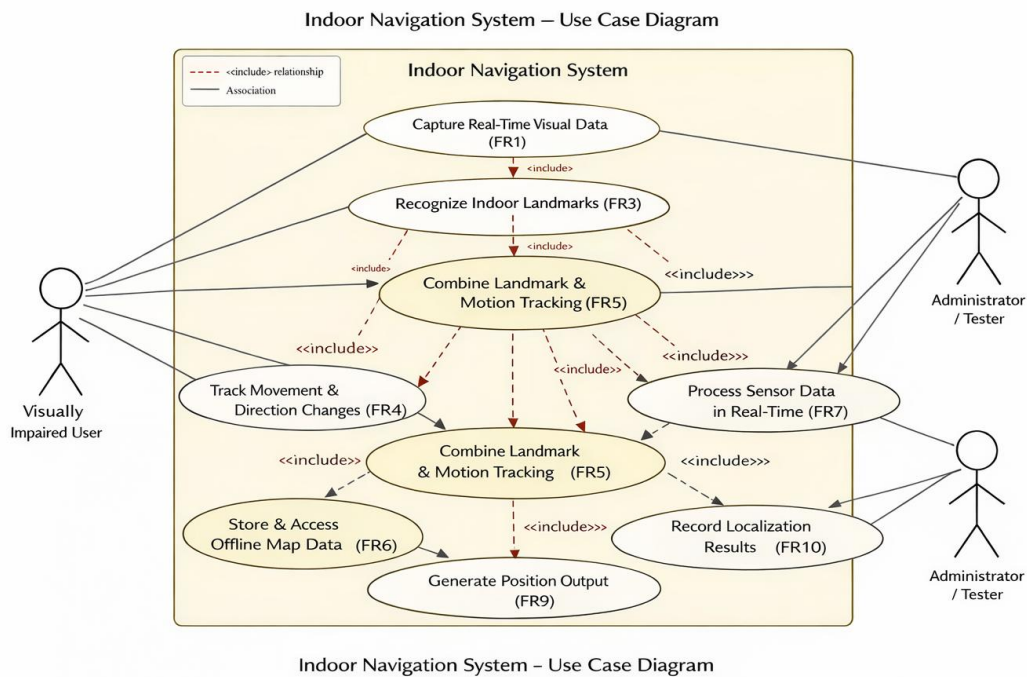


Figure 3 : User case diagram

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5.3 Non-Functional Requirements

The non-functional requirements define the quality attributes of the system.

Performance: The system should process visual and motion data in near real time and update the user's indoor position with minimal delay.

Usability: The system should be simple to use, require minimal user interaction, and support visually impaired users without increasing cognitive load.

1

Reliability: The system should consistently estimate indoor position and continue operation even when visual landmarks are temporarily unavailable.

Security and Privacy: The system should ensure that captured images, stored map data, and sensor logs do not expose personal or sensitive information unnecessarily.

Portability: The device should be lightweight, compact, and suitable for portable or wearable indoor use.

Offline Availability: The system should operate without continuous internet access and should not depend on cloud processing for core localization tasks.

Scalability: The system should allow future extension with additional sensors, new building maps, or connected guidance modules without major redesign.

Maintainability: The system should be modular so that the camera module, IMU module, localization engine, and offline map layer can be updated independently.

Energy Efficiency: The system should consume low power so that it can operate for a practical duration on portable battery power.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the proposed **Hybrid Indoor Positioning System with Offline Capability** is **visually impaired individuals** who need assistance with indoor localization and navigation. The proposed system will be especially useful for users who require support when moving inside **universities, hospitals, offices, shopping malls, public buildings, and rehabilitation centers**.

Potential customers include:

- Visually impaired people who need an affordable indoor positioning and navigation support system
- Hospitals, rehabilitation centers, and disability support organizations
- Universities, schools, and training institutions with visually impaired users
- Assistive technology organizations and accessibility programs
- Public and private institutions that aim to improve indoor accessibility

Customer Persona: The primary customer persona for the proposed system is a **visually impaired student, employee, or daily user** who needs a portable, reliable, and affordable assistive solution for identifying location and direction inside buildings without depending on constant internet connectivity.

6.2 Value Proposition

User Problem	Proposed Solution	Value to the User
Visually impaired individuals face difficulty navigating indoor environments safely.	Smart spectacles with camera, LiDAR, and IMU sensors for obstacle detection.	Improves safe navigation and reduces risk of collisions.
Traditional tools such as white cane cannot detect overhead obstacles or staircases.	Multi-sensor integration for detecting objects and elevation changes.	Provides more reliable hazard detection.
Many assistive navigation systems depend on internet connectivity.	Edge computing using Raspberry Pi for offline processing.	Allows reliable operation without internet access.
Smart commercial assistive devices are often expensive.	Low-cost hardware components used in the proposed system.	Provides an affordable assistive solution for visually impaired individuals.

Table 1 : Value Proposition of the Proposed Smart Spectacle System

6.3 Revenue Model

The proposed system can follow a **product-and-service-based revenue model**, where the indoor positioning device is sold directly to individual users or through assistive technology distributors and institutions.

Possible revenue sources include:

- Direct sales of the indoor positioning device
- Institutional deployment packages for hospitals, universities, and accessibility centers
- Partnerships with rehabilitation centers and disability support programs
- Government or NGO-supported accessibility initiatives
- Licensing the localization software or system design to assistive device manufacturers

6.4 Cost Estimation

The estimated prototype cost of the proposed system is based on the required hardware components.

Component	Estimated Cost (LKR)
Raspberry Pi Zero 2 W	8,890
Camera Module V2	8,000
LiDAR Sensor	1,550
IMU Sensor	680
Power System	1,700
Other Components	5,000
Estimated Prototype Cost	25,820 LKR

Table 2 : Estimated Prototype Cost of the Proposed Smart Spectacle System

6.5 Pricing Strategy

The pricing strategy of the proposed system will focus on **affordability, accessibility, and institutional adoption**.

Possible pricing approaches include:

- Affordable pricing model for visually impaired users in developing regions
- Subsidized pricing through government, NGO, or healthcare support programs
- Institutional package pricing for hospitals, universities, and accessibility centers
- Reduced pricing for bulk purchases or pilot deployments

The estimated retail price of the device can be around **55,000 – 65,000 LKR**, depending on final production cost, packaging, support, and deployment features.

6.6 Competitive Advantage

The proposed system has several competitive advantages compared to existing assistive navigation technologies.

- **Hybrid sensor integration** using camera-based landmark recognition and IMU-based motion tracking
- **Offline capability**, allowing indoor positioning without continuous internet connectivity
- **Infrastructure-light design**, reducing dependence on RFID tags, BLE beacons, or special building installations.
- **Edge-based processing** using Raspberry Pi for local and privacy-friendly operation.
- **Affordable and modular architecture** compared to more expensive commercial assistive systems.
- **Focused on indoor self-localization**, not only obstacle detection or object recognition.

6.7 Intellectual Property Considerations

The proposed system may include novel design and implementation elements related to **hybrid offline indoor positioning and assistive localization architecture**.

Potential intellectual property considerations include:

- **Patent protection** for the hybrid localization workflow or sensor fusion method, if it demonstrates novelty
- **Copyright protection** for the software, algorithms, and system implementation
- **Trademark protection** for the product name or brand identity in future commercialization
- **Possible licensing or collaboration** with assistive technology organizations or device manufacturers
- **Protection of system design documentation and deployment models** as part of product development assets

7. BUDGET AND JUSTIFICATION

7.1 Estimated Budget Breakdown

The estimated budget required to develop the prototype of the proposed **IoT-Based Spectacle for Indoor Navigation and Safety of Visually Impaired** is presented in Table 3.

Component	Quantity	Estimated Cost (LKR)
Raspberry Pi Zero 2 W	1	8,890
Raspberry Pi Camera Module V2	1	7,950
VL53L1X LiDAR Sensor	1	1,550
MPU6050 IMU Sensor	1	680
Li-Po Battery (3.7V 3000mAh)	1	1,550
MT3608 Boost Converter	1	150
MicroSD Card (32GB)	1	3,100
Earphone	1	810
Sunglass Frame and Wiring	1	1,000
Estimated Prototype Cost		~ 25,680 LKR

Table 3 : Estimated Budget Breakdown for Prototype Development

7.2 Hardware Cost Justification

The bulk of the budget is spent on hardware components that are necessary for the building of the prototype system.

- **Raspberry Pi Zero 2 W:** This will be used for the building of the embedded processing unit of the object detection system.
- **Raspberry Pi Camera Module V2:** This component will be used for the collection of data for the building of the project.
- **VL53L1X LiDAR Sensor:** This sensor will be used for the detection of distance for the detection of obstacles such as staircases.
- **MPU6050 IMU Sensor:** This sensor will be used for the detection of the orientation of the head, thus improving the accuracy of the navigation.
- **Li-Po Battery & Boost Converter:** This will be used to power the portable object detection system. It will also be used to power the raspberry pi.
- **MicroSD Card:** This will be used to store the operating system, software programs, and the machine learning model used by the object detection system.
- **Earphone:** This will be used for the provision of audio feedback to the user of the object detection system.
- **Sunglass Frame & Wiring:** This will be used for the connection of all the hardware components to the object detection system.

7.3 Software Requirements

The proposed system primarily utilizes **open-source software tools and libraries**, which reduce development and deployment cost. These tools support **computer vision, IMU sensor integration, sensor fusion, offline data handling, and real-time indoor positioning** for the proposed Hybrid Indoor Positioning System with Offline Capability.

Software Tool	License Cost
Python Programming Language	Free
OpenCV Library	Free
Raspberry Pi OS	Free
NumPy	Free
IMU Sensor Library (e.g., Raspberry Pi OS MPU6050 library)	Free

Table 4 : Software Tools Used in the Proposed System

7.4 Cloud Services

The proposed system is designed to operate mainly using **edge computing**, where most data processing occurs locally on the Raspberry Pi device.

This design reduces dependence on cloud computing services and avoids additional operational costs. Therefore, no cloud service expenses are required for the prototype development stage.

7.5 Human Effort Cost

The development of the system will be carried out by the **research team members as part of their academic research project**. Therefore, no direct financial cost is allocated for human **resources** during the prototype development stage.

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 Task Breakdown per Member

Member	System Component	Key Tasks
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Member 1	Intelligent Vision-Based Danger Detector 	Task 1: Collect and prepare image dataset for indoor hazards. Task 2: Configure YOLO object detection model for edge devices. Task 3: Train and optimize the model using labeled datasets. Task 4: Evaluate detection accuracy and performance. Task 5: Integrate detection system with the smart spectacle platform.
Member 2	Intelligent Security & Remote Guardian System 	Task 1: Design and develop the guardian mobile application. Task 2: Implement fall detection and inactivity monitoring algorithms. Task 3: Develop geo-fencing functionality for safe zones. Task 4: Integrate cloud database for real-time monitoring. Task 5: Implement SOS emergency alert system.
Member 3 (ME)	Smart Spectacle Hardware Module with Multi-Sensory Integration 	Task 1: Design the wearable smart spectacle hardware structure. Task 2: Integrate Raspberry Pi with camera, LiDAR, and IMU sensors. Task 3: Implement sensor data acquisition and synchronization. Task 4: Develop audio output interface for navigation alerts. Task 5: Implement push-button control for system interaction.

Member 4	Hybrid Indoor Positioning System with Offline Capability 	Task 1: Develop indoor localization algorithm using IMU sensors. Task 2: Implement vision-based landmark recognition. Task 3: Integrate dead-reckoning for indoor positioning. Task 4: Combine vision and IMU data for hybrid localization. Task 5: Integrate positioning system with navigation and hazard detection modules.
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Table 5 : Task Breakdown per Member for the Proposed Smart Spectacle System

8.2 Timeline Alignment

The development process of the proposed IoT-Based Spectacle for Indoor Navigation and Safety of Visually Impaired will be carried out in an aligned manner with an appropriate timeline so that it can be achieved systematically during the entire research period. The project will be achieved through different key milestones that will be included in the research process.

The project will be aligned with the following timeline:

- Phase 1 – Literature Review and Problem Analysis (Month 1):**
 Thorough research will be performed based on the existing research works on assistive technologies, indoor navigation systems, computer vision models, and IoT-based wearable devices.
- Phase 2 – System Architecture and Design (Month 2):**
 Design of the entire system, including hardware as well as software components, shall be conducted.
- Phase 3 – Hardware and Software Development (Month 3-4):**
 The development of the individual components of the entire system will be performed.
- Phase 4 – System Integration and Testing (Month 5):**

All the components of the entire system will be integrated and the testing of the entire system will be performed.

- **Phase 5 – Evaluation and Documentation (Month 6):**
Analysis of the entire system, completion of the research report, and completion of the project's presentation.

8.3 Balanced Workload Distribution

The project workload is distributed among the four members of the group based on the technical expertise of each member and the needs of the proposed project. Each member of the group is responsible for the development of a particular module of the project, including vision-based hazard detection, the remote guardian monitoring system, the smart spectacle hardware module, and the hybrid indoor positioning system. This ensures the development of all the major parts of the project efficiently, with all the members of the group participating equally.

Besides the development of the project's individual parts, some activities such as integration, testing, and report writing will be undertaken jointly by all the members of the group to ensure the equal participation of all members in the successful completion of the research project.

9. GANTT CHART

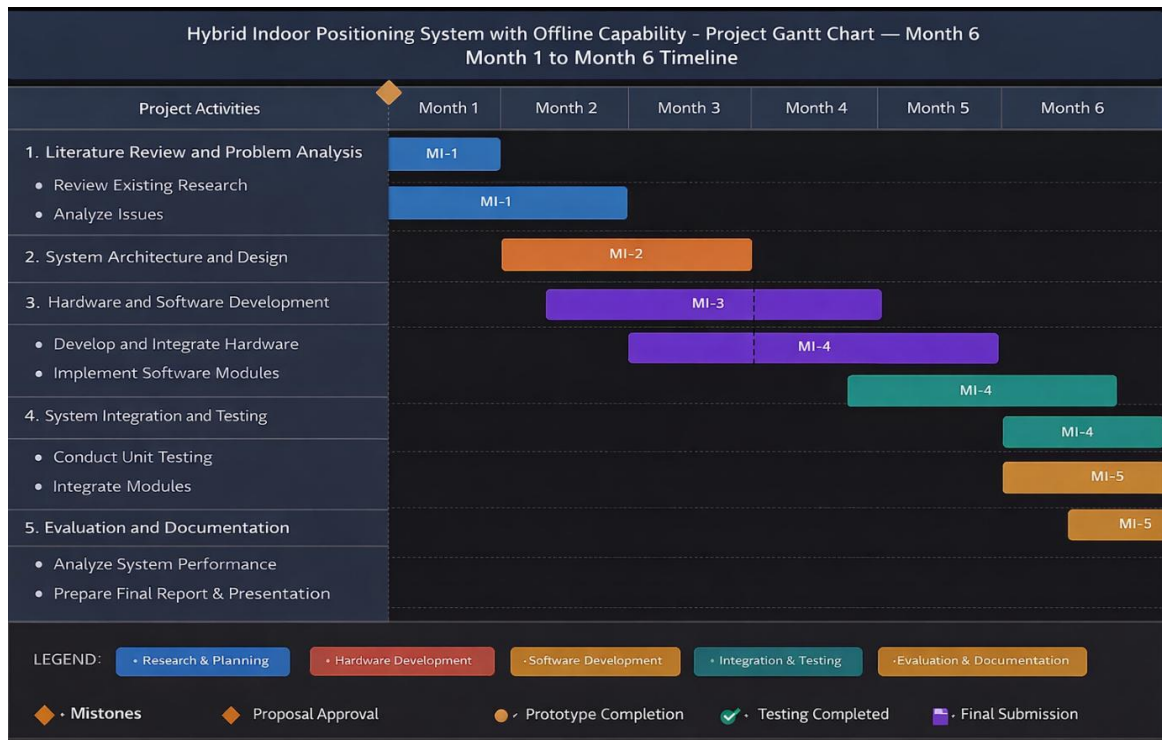


Figure 4 : Gantt Chart for the Proposed Smart Spectacle Research Project

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11. APPENDICES

Appendix A – Survey Instruments

The following survey questionnaire is designed to understand the **challenges faced by visually impaired individuals when navigating indoor environments** and to evaluate the potential usefulness of a **Hybrid Indoor Positioning System with Offline Capability**.

The survey focuses on understanding user needs related to **indoor localization, navigation assistance, and offline positioning technologies** that can support independent mobility inside buildings such as hospitals, universities, offices, and public facilities.

Target Audience

The target audience for the survey includes the following groups:

- **Visually impaired individuals** who experience difficulties navigating indoor environments
- **Caregivers or family members** of visually impaired persons
- **Professionals in assistive technology**, rehabilitation, or accessibility services
- **Researchers or developers** working on navigation and assistive technologies

Survey Questions

1. What type of difficulties do visually impaired individuals commonly face when navigating indoor environments?

- Difficulty identifying their current location inside buildings
- Difficulty determining walking direction or orientation
- Difficulty recognizing corridors, doors, or junctions
- All of the above

2. Which assistive tools are commonly used by visually impaired individuals for mobility and navigation?

- White cane
- Guide dog

- Smartphone navigation applications
 - Electronic assistive devices
3. How important is accurate indoor positioning or location awareness for visually impaired users?
- Very important
 - Important
 - Moderately important
 - Not important
4. What type of navigation information would be most useful in an indoor positioning system?
- Current location inside the building
 - Direction of movement or heading
 - Identification of nearby landmarks (doors, corridors, staircases)
 - Combination of all the above
5. How useful would an offline indoor positioning system be for visually impaired users in environments where internet connectivity is limited?
- Very useful
 - Useful
 - Slightly useful
 - Not useful
6. Which feature would be most important in a smart indoor positioning system?
- Real-time indoor location identification
 - Direction and movement tracking
 - Landmark recognition (doors, corridors, room signs)

- Offline navigation capability

7. How important is it for assistive navigation systems to work without internet connectivity?

- Very important
- Important
- Moderately important
- Not important

8. Which type of device would be most convenient for an indoor navigation solution?

- Wearable device
- Smartphone-based system
- Portable handheld device
- Smart assistive glasses

9. In which indoor environments would such a navigation system be most helpful?

- Hospitals and healthcare facilities
- Universities and educational institutions
- Shopping malls and public buildings
- Office buildings and workplaces

10. Would you be willing to use an assistive system that helps determine your location and direction inside buildings?

- Yes
- Maybe
- No

Appendix B - Detailed System Diagrams

- **Smart spectacle hardware architecture diagram**

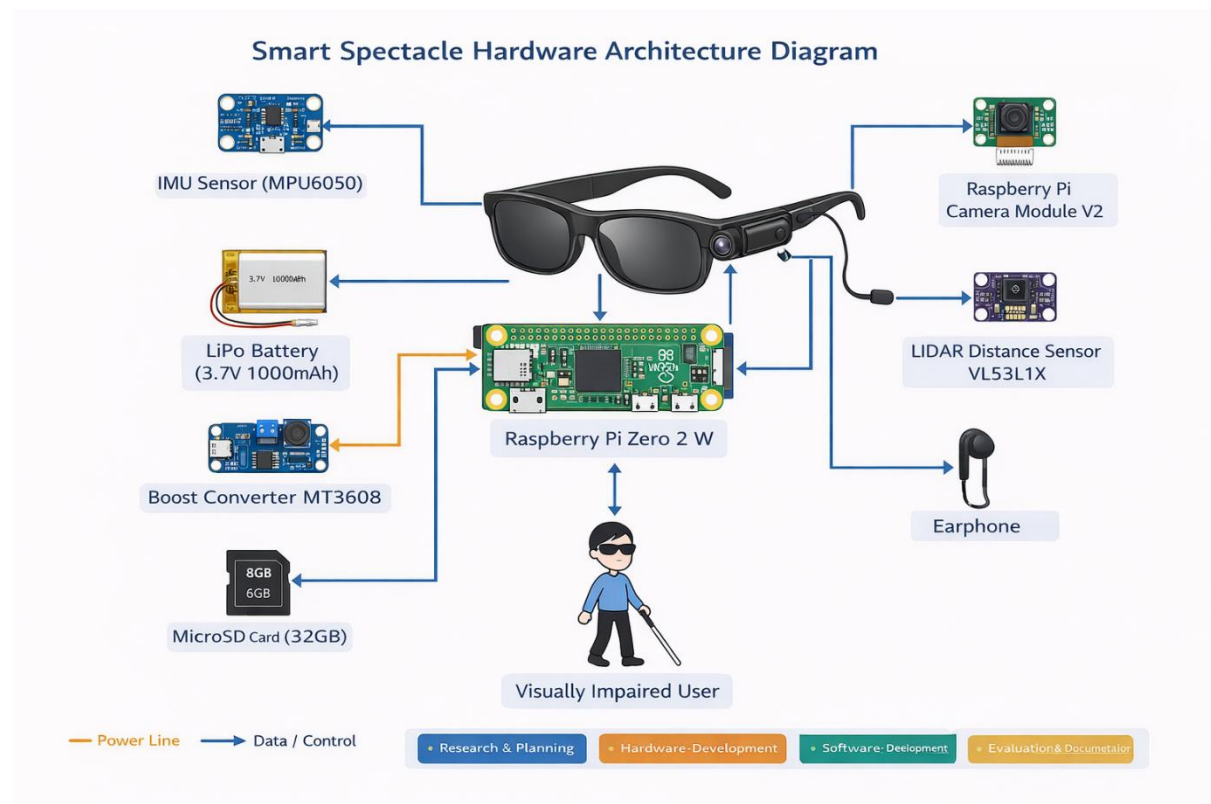


Figure 5 : Smart Spectacle Hardware Architecture Diagram

- Data processing pipeline

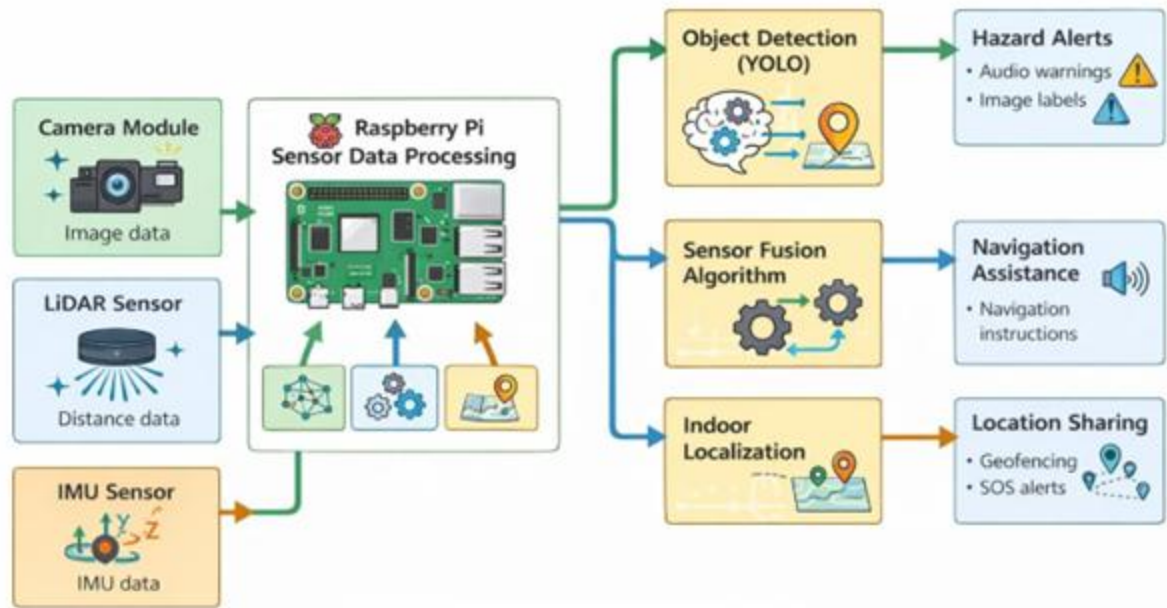


Figure 6 : Sensor Data Processing Pipeline

Appendix C – Preliminary UI Mockups

This appendix presents the preliminary user interface designs for the Guardian Monitoring Mobile Application, which allows guardians to monitor the safety and location of visually impaired users using the proposed smart spectacle system.

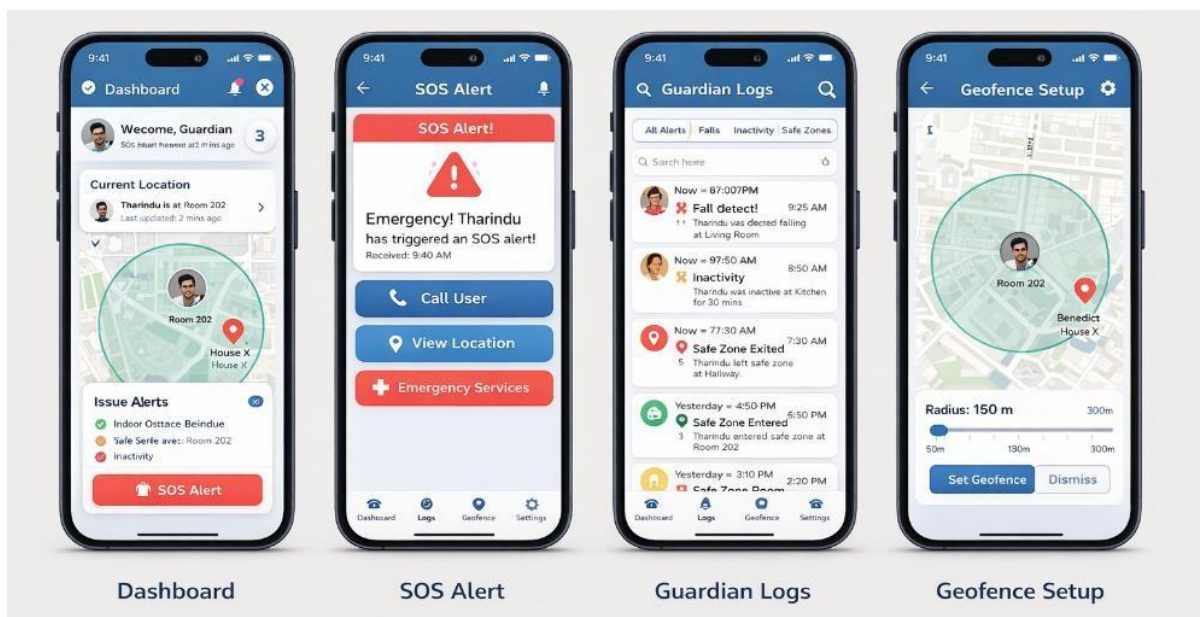


Figure 7 : Preliminary User Interface Mockups for the Guardian Monitoring Mobile Application

Appendix D – Risk Analysis

Risk Factor	Impact	Probability	Mitigation Strategy
Hardware component failure (Raspberry Pi, Camera, or IMU sensor)	The indoor positioning system may stop functioning or provide incorrect localization results	Medium	Use reliable hardware modules, ensure proper wiring, and perform regular hardware testing and maintenance
IMU sensor drift or motion tracking errors	Incorrect estimation of user movement, direction, or turning patterns	Medium	Apply motion filtering techniques and combine IMU data with visual landmark recognition (sensor fusion)
Camera performance affected by lighting conditions	Landmark recognition accuracy may decrease in low-light or poor visibility environments	Medium	Use image preprocessing techniques and combine vision data with IMU-based motion tracking
Incorrect or incomplete landmark recognition	The system may fail to correctly identify indoor landmarks such as doors or corridors	Medium	Improve landmark detection algorithms and update the local landmark database regularly
Offline map or landmark database errors	Localization accuracy may decrease if map data is outdated or incomplete	Low	Maintain and update local map datasets and verify building landmarks during testing
Battery power limitations	The embedded system may shut down during operation	Medium	Use high-capacity rechargeable batteries and optimize system power consumption

Table 6 : Risk Analysis for the Smart Spectacle Hardware Module (Component 3)

Appendix E – Ethical Considerations

This research is carried out following the guidelines on research ethics provided by the Sri Lanka Institute of Information Technology. This institute has an Ethics Review Committee (ERC), which deals with the review of the research proposal and ensures that all the research activities are carried out ethically.

The following ethical principles are considered in this research project:

- **Personal Data:**
This research does not require any personal data from any individual. If any data is required in the experiment for the system, it is related to objects in the environment, such as furniture, doors, staircase, etc.
- **Privacy protection:**
If images are captured during experiments in indoor environments, they will not intentionally include identifiable individuals. Any accidental appearance of people will not be stored or used for analysis.
- **Responsible use of technology:**
The developed smart spectacle system is only intended to assist and enhance the safety and mobility of people with visual impairments.
- **Compliance with institutional ethics guidelines:**
This research has been conducted based on the ethical guidelines suggested by the SLIIT ERC to carry out ethical academic research.
- **Academic integrity:**
All the information sources, data sets, and research material used in the current study are properly credited to ensure academic integrity.

SLIIT Ethics Review Committee:

<https://www.sliit.lk/about/about-sliit/ethics-review-committee>